

STEELE L. Z. FRITCHIE

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Electrical engineering student with a strong background in designing wired circuits using the Multisim program, 3D modeling using programs such as Autodesk Inventor, SolidWorks, and Onshape, and file and project management who is eager to contribute engineering and critical thinking skills to an organization that serves our country.

LIBERTY UNIVERSITY

B.S. Electrical Engineering

Liberty University, Lynchburg, VA

Graduation May 2028

GPA: 2.98

LIBERTY ROCKETRY - International Rocketry Engineering Competition (IREC)

Lead Airbrakes Engineer

June 2026 - Present

- Leading a team of five elite engineers, conceptualize and develop an airbrake system to use on the 2027 rocket, as well as leverage understanding of systems to assist the overall configuration of the ballast and other systems.
- Petal-Actuated flaps create a drag producing surface to hit a precise selected apogee
- Using real-time input variables, a machine learning model trained on 100,000+ simulated flights predict a projected apogee and calculates optimal deployment and retraction time and length of airbrake flaps.

Lead Avionics Engineer

August 2026-June 2026

- Led a team of 16 engineers, using knowledge and understanding of rocketry to alter, optimize, and adjust existing and integrating new systems to the OMEGA 2026 LU rocket, which placed 18th at 2026 IREC.
- Optimize a LORA mesh network by reducing nodes, shifting components, and using YAGI antennas to combat interference experienced at competition when relaying information including start commands and packets.
- Reworked ground station code which analyzes data from sensors on a custom student developed STM-32 based flight computer which relays real time data to graph on a student developed ground station program.
- Iterative design processes used to integrate electronic components for critical recovery systems as well as telemetry through extensive 3D modeling and printing of mounting solutions for electronics and integrating them on the rocket.

Avionics Design Engineer

September 2024 – June 2025

- Used iterative design processes to 3D model and print an optimized avionics bay which holds mission-critical electronics.
- Created, wired, and tested several versions; the optimized avionics bay was used in the team's 10k COTS 2025 competition rocket, Trinity, and received several on-the-spot commendations from IREC judges for its design.
- Featured wire channels and component slots where every component and wire had an intentional placement and routing.

PROJECT LEAD THE WAY - Highschool Tech Center

Project Lead The Way (PLTW) Engineering Program

February 2023 – April 2024

Carroll County Career and Technology Center, Westminster, MD

GPA: 4.0

BRS Trucking Solutions (PLTW Capstone Project)

(PLTW) February 2024 – May 2024

- Invented a solution to minimize accidents involving large trucks. Researched prior solutions and patents, defined requirements and specifications, and contacted local experts to gain insight. Developed in Inventor 3D modeling software. Prototype passed all identified testing criteria. Market share and future development were analyzed.

Date In History

(PLTW) April 2024

- Used knowledge of circuit principles to create a truth table, Karnaugh maps, and the Multisim simulation program to design a circuit which used a three-input system and wire a physical breadboard with the designed circuit.

Automated Guided Vehicle (AGV)

(PLTW) February 2024

- Designed, built, and programmed an AGV which can follow, without human input, a variable path of colored tape on the ground (a left color, a middle color, and a right color) which consists of turns up to ninety degrees and stops at several points indicated by blue tape before advancing on the path.

Reverse Engineering Project

(PLTW) January 2024 – February 2024

- Conducted research on existing products and patents to develop design specifications and concept sketches for a phone stand. After selecting a final design, property and stress analysis were performed on a 3D model of the phone stand. When the model was printed, evaluation and reflections were done on the process and final product.

TECHNICAL SKILLS

Electrical: PCB Design, Circuit Simplification and Logic Design, Breadboarding and Prototyping, Karnaugh Maps

Radio Frequency: LORA Mesh Networking, Spreading Factor Control, Antenna Systems, Radio Frequency Interference Analysis

Computer Aided Design: SolidWorks, Autodesk Inventor, Onshape, 3D Modeling Printing and Integration, Iterative Design

Simulation and Analysis: Multisim, RocketPy, OpenRocket, Finite Element Analysis, Computational Fluid Dynamics, Circuit Analysis

Coding / Software: Python, C++, Machine Learning/Predictive Modeling, Ground Station Software, Github, Code/Software Analysis